When the Attacker Is Indistinguishable from the Weather: A Cross-Layer Benchmark and a Unified Perturbation Budget for Certified UAV Autonomy

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***Abstract*—A drone cannot tell from its navigation inputs whether a growing position error comes from a jammer, a drifting sensor or wind. An attacker can therefore disguise interference as a tolerated benign disturbance, so mission guarantees must not rely on detection. We formalise this *adversarial equivalence* with a total-variation bound on detection advantage and show that certificates gated on detector silence can be violated.**

A *unified perturbation budget* over adversarial, GNSS, sensor-fault and environmental terms bounds the probability of undetected mission failure. Dependence-sharp aggregation (Makarov bounds, Rearrangement Algorithm) makes it hold for every coupling of benign sources, even one set by an attacker who controls when jamming happens. A cross-layer benchmark pairs each network or environmental event with the navigation window it induces, under a mandatory provenance flag.

In 1,909,600 simulated missions, 35 optimised attackers in five disguise families evade four detectors in at least 87% of missions. A trusted re-anchoring alarm limits evasions to a few metres; without it, undetected deviation reaches 9.9–73.9 m. The bound held at all 270 pilot points; stronger optimisers over 1,620 points found 5 violations, none from disguise, all from two broken theorem conditions (an unchecked actuator limit, a small calibration sample). Once fixed, 1,296 fresh-seed tests showed 0 violations, though the empirical stealth budget was again exceeded and a provable one is 5.8× larger; measured jamming data halve the certified floor at 10 m. UAV safety cases should therefore budget for attacks disguised as tolerated disturbances, not trust detection to catch them.

I. Introduction

An autonomous UAV is exposed to an adversary at several interfaces at once. Commands arrive over a command-and-control radio link, peers exchange position reports over a multi-hop mesh, the flight controller talks to its motor controllers over an on-board bus, and the navigation filter fuses satellite, inertial and occasionally visual measurements into the state estimate on which the control loop acts. Each interface has its own defensive literature: network intrusion detection systems (IDSs) for the links and the bus [1]–[3], physical-invariant and residual monitors for the estimator [4]–[6], and certified robustness for learned components [7], [8]. Our concern is a question none of these defences is designed to answer. When some attacks inevitably get through, can the mission still be completed, and can that be proved?

A companion paper [9] answered a narrow version of this question for a single attack class. A relay that delays peer position reports by just less than a detector's threshold produces a bounded staleness, the staleness produces a bounded navigation error, and a Lipschitz–Grönwall tube converts that error into a certified floor on the mission-completion rate (MCR). That chain assumed the attacker announces itself as an attacker: its perturbation was the only one in the budget, and its type was known.

a) The attacker who dresses as the weather.: Real navigation errors have many causes. A GNSS receiver drifts with the ionosphere and multipath; an accelerometer bias steps or random-walks; wind pushes the airframe, and the controller fights back. A defender therefore tunes every monitor to tolerate these benign disturbances, and the tolerance is precisely the space in which an attacker can hide. Dash et al. observed that control-invariant monitors "have to accommodate some degree of deviation … due to environmental factors such as friction or wind" [6]. Roy and Dey proved that an actuator attack lying in the range of the fault-input matrix is indistinguishable from a fault [10]. Khazraei et al. showed that a GPS-only attacker can cause arbitrarily large deviation against *any* detector when the closed loop is incrementally exponentially stable [11], and a recent spoofer deliberately exploited the controller's wind compensation to deviate a PX4 vehicle undetected [12]. We take these observations to their security conclusion:

> *From the navigation input alone, shaped adversarial interference, a sensor fault and an environmental disturbance can be indistinguishable. An attacker will therefore choose the disguise the defender models least well. A certificate that is sound only against a declared attack model is unsound against the attacker who dresses as the weather.*

The conclusion has two consequences that the single-source view misses. First, a detector tuned to malicious signatures is the wrong last line of defence: the decisive attack is the one that looks benign. Second, a guarantee is worth having only if it is *source-agnostic*, meaning it bounds the mission outcome

whatever mixture of attack, fault, outage and wind produced the perturbation, and whatever their mutual dependence.

b) Contributions.:

- **A formal threat model of adversarial equivalence (Section III).** We define ε-observational equivalence between an attack and a class of benign disturbances, bound any detector's advantage by the total-variation (TV) distance (Lemma 1), and show that a detector-gated certificate can be violated by an equivalent attack whose undetected deviation exceeds the corridor (Proposition 1). The observation that attacks can mimic faults is not new [10], [13], [14]; what is new is making the disguise an optimisation variable of the attacker and deriving from it a requirement on the certificate.
- **A unified perturbation budget with mixed semantics (Section IV).** The budget $\delta_{\text{tot}} = \delta_{\text{net}}(\theta) \oplus \delta_{\text{adv}}(\theta) \oplus \delta_{\text{gnss}} \oplus \delta_{\text{fault}} \oplus \delta_{\text{env}}$ combines deterministic worst-case terms, stochastic terms and a bounded-unknown term represented as a probability box (p-box). We prove a dependence-sharp bound on the probability that a mission fails undetected (Theorem 1) and characterise its tightness (Proposition 2). The union bound is the classical standard bound [15], [16]. It is sharp for two sources, so it is the right quantity when the coupling is unknown. Against a *known* benign coupling it is loose: by a factor up to the number of sources under comonotonicity, and without bound under independence. Deterministic terms add no dependence gap at all. This corrects the hypothesis in our earlier proposal, that a joint treatment recovers the slack precisely under comonotonicity.
- **A cross-layer benchmark with enforced provenance (Section V).** Six network and navigation corpora are ingested through lossless adapters, and fault and environment sources are added under a schema whose unit of record is the cross-layer pairing. Every pairing states whether its coupling was measured on one airframe, derived from a physical model, or aligned in simulation.
- **An evasion study and a soundness test (Sections VI and VII).** Against four detector designs and their combination, an optimising attacker searches over five disguise families. We cross-evaluate every optimised attacker against every detector. We then measure how much undetected deviation each detector admits per disguise, how the stealth budget scales with the re-anchoring interval, and where each aggregation rule loses tightness. Finally, we test whether the bound holds on held-out attackers under Holm-corrected tests, with negative controls that demonstrate the test's power.
- **A negative result and its repair (Sections VII-E and VII-F).** Stronger optimisers and a test scaled to 1,620 points found the pilot instantiation unsound: 5 Holm-significant violations, traced to two conditions that it assumed but did not secure. Neither concerns disguise. We repair both with a run-time saturation alarm and a finite-sample p-box, and judge the repair on fresh seeds. A linear-programming bound on the stealth budget, valid for every attack, exists only once the saturation alarm is armed, and it is $5.8\times$ the empirical one.

c) Scope.: The pilot evaluation in this paper uses a reduced-order lateral model of a multirotor with a Kalman filter (KF) rather than PX4's EKF2 on a full airframe. Section VIII and the released runbook state precisely which results must be re-established in software-in-the-loop (SITL) simulation before they are claimed for a deployed autopilot. The adversarial term we can use is an empirical supremum over optimised attackers, not a proof. The provable bound we derive is too loose to certify useful margins. We test the empirical term against independently seeded attackers and report every exceedance and every violation.

II. Preliminaries and Problem Statement

A. Navigation loop and mission completion

Let $x(t) \in \mathbb{R}^n$ be the navigation state of the aircraft and $\dot{x} = F(x,u)$ its closed-loop dynamics under feedback u computed from the *estimated* state $\hat{x}$. A mission ω of horizon T follows a nominal path x^{nom} inside a corridor of half-width (margin) m. Writing $e(t)$ for the cross-track deviation of the true position from the nominal path, the mission completes if and only if

$$\sup_{t\in[0,T]} |e(t)| \le m. \tag{1}$$

The mission-completion rate is $\text{MCR} = \Pr_{\omega\sim\mathscr{M}}[(1)]$ for a stated mission distribution $\mathscr{M}$; it is a population quantity, not a statement about any single flight [17].

a) The parent chain.: If F is L-Lipschitz on the operating region (condition C1) and the navigation input is perturbed by at most δ, Grönwall's inequality [18] bounds the deviation over a certification horizon T_c by the tube radius $\rho = \delta e^{LT_c}$. For the relay-delay attacker, the companion paper [9] proved a residual-budget lemma: an attacker below a detector threshold θ on H malicious hops hides at most $\Delta(\theta) \le H\min(\theta, s)$ of staleness, with the contact-window slack s measured at 4.84 s over 15,392 relay hops. It also calibrated the delay-to-position rate on real jamming and spoofing flights at the supremum $\gamma = 1.625$ m/s. The network contribution to the deviation is therefore at most

$$e^{LT_c}\,\delta_{\text{net}}(\theta), \qquad \delta_{\text{net}}(\theta) = \gamma H \min(\theta, s), \tag{2}$$

with the measured local Lipschitz constant $L = 1.181$ and $T_c = 1$ s. At the paper operating point ($\theta = 0.25$ s, $H = 2$) this contributes 2.65 m.

B. Perturbation sources and their semantics

Five sources perturb the navigation loop. They differ not only in magnitude but in the *kind of statement* that can be made about them:

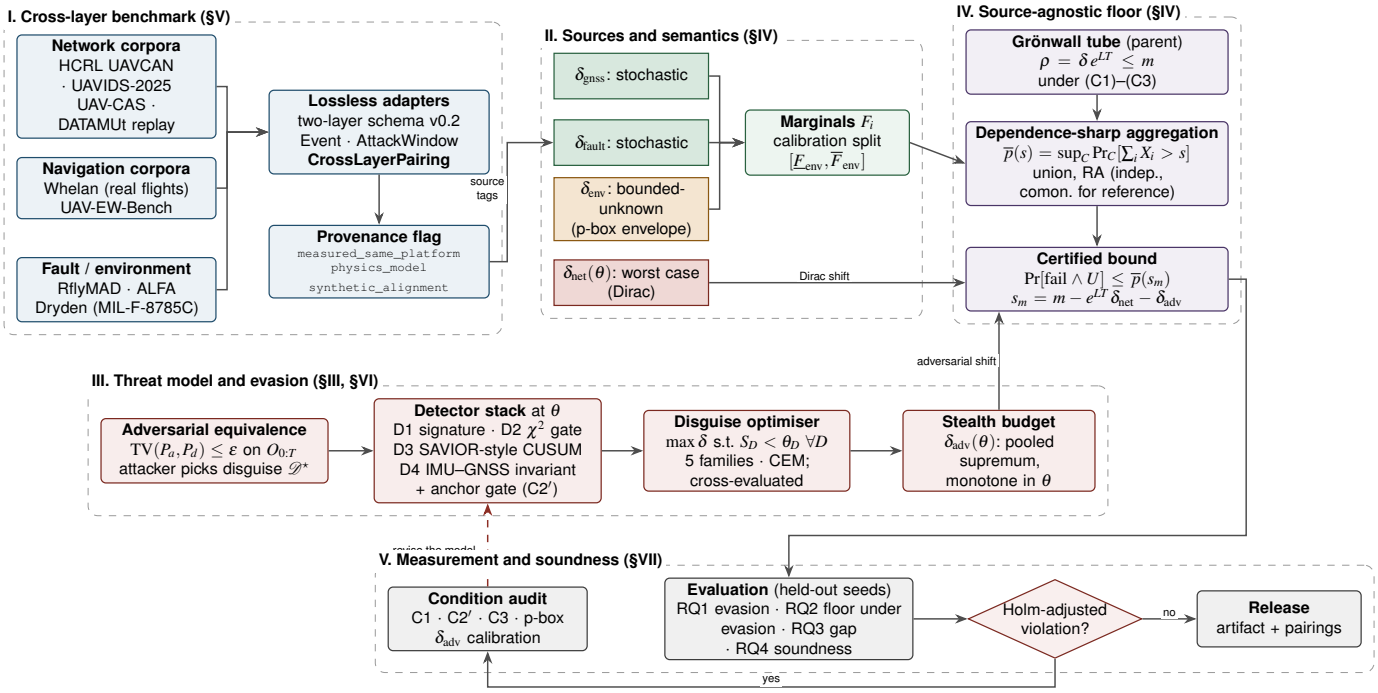


Fig. 1. Methodological framework. Phase I unifies network, navigation, fault and environment corpora under one schema whose unit of record is the cross-layer pairing, each carrying a mandatory provenance flag. Phase II assigns every perturbation source its probabilistic semantics and estimates its marginal on a calibration split. Phase III formalises adversarial equivalence and searches, for each detector stack and operating point θ, for the disguise that maximises undetected deviation; the pooled supremum is the stealth budget $\delta_{\text{adv}}(\theta)$. Phase IV aggregates the stochastic and bounded-unknown terms with dependence-sharp bounds and shifts the result by the deterministic terms, yielding a certified floor that does not depend on which source produced the perturbation. Phase V tests that floor on held-out attackers and rollouts; a Holm-significant violation triggers an audit of the stated conditions rather than a re-tuning of constants (dashed feedback).

Term	Origin	Semantics
$\delta_{\text{net}}(\theta)$	relay delay below θ	deterministic worst case
$\delta_{\text{adv}}(\theta)$	disguised GNSS offset below the detector stack at θ	deterministic worst case (stealth budget)
δ_{gnss}	noise, Gauss–Markov drift, outages, jamming noise	stochastic, known marginal
δ_{fault}	inertial bias steps and drift, actuator loss of effectiveness	stochastic, known marginal
δ_{env}	turbulence within a declared intensity range	bounded-unknown (p-box)

A deterministic term is a number that holds for every admissible behaviour. A stochastic term has a cumulative distribution function (CDF) that we estimate on a calibration split. A bounded-unknown term has a distribution known only to lie between a lower and an upper CDF, a probability box [19]. For wind, the box is spanned by the Dryden turbulence model of MIL-F-8785C [20] between light and moderate intensity. What the sources do *not* come with is a joint distribution. Nature couples gusts and outages in ways nobody has measured, and an attacker who induces jamming can choose when GNSS degradation coincides with gusts.

C. Problem statement

Given (i) the marginal laws of the stochastic sources, (ii) an envelope for the bounded-unknown source, (iii) a detector stack operating at false-positive rate (FPR) budget α with thresholds θ, and (iv) an attacker free to choose both the shape of its interference and its coupling to the benign sources, we seek:

- **P1** an upper bound on the probability that a mission fails *and* its attack goes undetected, valid for every coupling of the benign sources and every attacker, i.e., a *source-agnostic* bound;
- **P2** a characterisation of how much tightness each aggregation rule sacrifices and where the slack comes from;
- **P3** a measurement substrate that keeps the evidence behind every coupling explicit, and an evasion study that measures which disguise defeats which detector.

These define six research questions. RQ1: can an attacker mimic a fault, a GNSS drift or a gust well enough to evade a detector while still producing a large deviation, and which disguise does each detector miss? RQ2: does the source-agnostic floor bound the outcome of missions whose attack went undetected? RQ3: how loose are the union bound and the dependence-sharp bound in practice, and is the looseness due to dependence or to time? RQ4: is the floor sound at every held-out operating point under Holm-corrected tests, and does the test detect unsoundness when a condition is deliberately broken? RQ5: does the answer to RQ4 survive stronger attackers and a larger test, and if not, which condition fails? RQ6: how much does the floor change when the simulated GNSS marginal is replaced by measured jamming data?

Figure 1 summarises the methodology through which the paper answers these questions.

III. Threat Model: Adversarial Equivalence

A. Capabilities, knowledge and goal

The attacker can (a) delay relayed messages on up to H hops of the mesh, as in the companion paper, and (b) add an offset $d(t)$ to the GNSS position the navigation filter receives. Capability (b) is realised in practice by signal-level spoofing [21]–[23]; in this paper it is exercised only through simulator interfaces. The attacker cannot corrupt the inertial measurement unit (IMU), and it cannot corrupt a trusted re-anchoring fix, such as a visual landmark or an authenticated position broadcast, that arrives at least once every T_a seconds (condition C2′, which generalises the parent's re-anchoring condition from the network channel to the GNSS channel). We return to IMU injection [24], [25] in Section VIII.

Following Kerckhoffs's principle, the attacker knows the detector designs and their thresholds. It also observes the local wind, which a co-located attacker can measure. It does not know the realisation of the benign noise. Its goal is to maximise the vehicle's cross-track deviation while raising no alarm at any time during the mission. Being detected late counts as being detected.

B. Observational equivalence

Let $O_{0:T}$ denote everything the defender observes: GNSS fixes, IMU samples, commanded accelerations and anchor fixes. Write P_a for the law of $O_{0:T}$ under attack a (with the benign sources present), and P_d for its law under a benign disturbance d drawn from a class $\mathscr{D}$, such as inertial-bias drift, GNSS multipath drift, or turbulence of a given intensity.

Definition 1 (ε-observational equivalence)**.** *Attack a is ε-equivalent to the class $\mathscr{D}$ if* $\inf_{d\in\mathscr{D}} \mathrm{TV}(P_a,P_d) \le \varepsilon$, *where* $\mathrm{TV}(P,Q) = \sup_A |P(A)-Q(A)|$ *over measurable events A.*

Lemma 1 (Detection advantage; Le Cam)**.** *For every detector $\psi : O_{0:T} \to \{0,1\}$ and every $d \in \mathscr{D}$,* $\Pr_a[\psi=1] \le \Pr_d[\psi=1] + \mathrm{TV}(P_a,P_d) \le \Pr_d[\psi=1] + \sqrt{\mathrm{KL}(P_a\|P_d)/2}$.

Proof. The event $\{\psi=1\}$ is measurable, so $|P_a(\psi=1) - P_d(\psi=1)| \le \mathrm{TV}(P_a,P_d)$ by definition. The second inequality is Pinsker's [26]. □

Lemma 1 unifies three earlier notions. The "hidden" attacks of Hashemi et al. raise alarms at the false-alarm rate [14]; the ε-stealthy attacks of Bai et al. bound the Kullback–Leibler (KL) divergence [13]; and the fault-mimicking attacks of Roy and Dey are the noiseless case $\varepsilon = 0$ in which $\mathscr{D}$ is a fault class [10]. A detector with false-alarm rate α on $\mathscr{D}$ thus catches an ε-equivalent attack with probability at most $\alpha+\varepsilon$, whatever its design.

Definition 2 (Disguise-optimal attacker)**.** *For a detector stack $\Psi_\theta = \{\psi_{D,\theta_D}\}_D$ and disguise families $\{\mathscr{D}_k\}$, each parameterised by $\phi \in \Phi_k$, the attacker solves*

$$\delta_k^\star(\theta) = \sup_{\phi\in\Phi_k} \mathbb{E}\big[\Delta_\phi \cdot \mathbb{1}\{\psi_{D,\theta_D}(O^\phi)=0 \ \forall D\}\big], \qquad (3)$$

where $\Delta_\phi = \sup_t |e(t)|$ is the induced deviation. It then selects $k^\star(\theta) = \arg\max_k \delta_k^\star(\theta)$. Dressing as the weather means choosing $k^\star$ to be the class the stack tolerates most.

Proposition 1 (Detector-gated certificates and equivalence)**.** *Let ψ have false-alarm probability at most α under every $d \in \mathscr{D}$, and let $f_{\mathscr{D}}$ be a floor valid under $\mathscr{D}$. Let a be ε-equivalent to $\mathscr{D}$ with $\alpha+\varepsilon<1$, and let $q_a = \Pr_a[\mathit{fail} \mid \psi=0]$. Then*

$$\mathrm{MCR}_a \ \le\ 1 - q_a(1-\alpha-\varepsilon).$$

Hence $f_{\mathscr{D}}$ is violated by a whenever $q_a(1-\alpha-\varepsilon) > 1-f_{\mathscr{D}}$. A floor that is to remain valid against every such attack must therefore bound the deviation the attack itself causes on undetected missions; the detector's silence is no evidence of benignity.

Proof. By Lemma 1, $\Pr_a[\psi=0] \ge \Pr_d[\psi=0] - \varepsilon \ge 1-\alpha-\varepsilon$. Then $1-\mathrm{MCR}_a \ge \Pr_a[\mathrm{fail} \wedge \psi=0] = q_a \Pr_a[\psi=0]$. □

The proposition is the formal content of "the decisive attack is the one that looks benign". Without condition C2′, Khazraei et al. show that a GPS-only attacker can make the undetected deviation grow without bound in an incrementally exponentially stable closed loop [11], so q_a approaches one for any fixed corridor. With C2′, the deviation an undetected attacker can accumulate is limited by the anchor interval. The stealth budget $\delta_{\mathrm{adv}}(\theta)$ of Section IV is our estimate of that limit.

IV. The Unified Budget and the Source-Agnostic Floor

A. From sources to an additive budget

For each benign source $i \in \{\mathrm{gnss}, \mathrm{fault}, \mathrm{env}\}$, let $e_i(t)$ be the deviation the closed loop would exhibit under source i alone, and define its *sup-deviation* $X_i = \sup_t |e_i(t)|$. Let e_{adv} be the deviation under the attack alone.

Lemma 2 (Sub-additivity for additive inputs)**.** *Fix the time-varying parameters of a mission (GNSS outage schedule, actuator effectiveness, anchor schedule). If the closed loop is then linear in its additive inputs (wind force, sensor noise and bias, attack offset), the deviation decomposes as $e = e_{\mathrm{ben}} + e_{\mathrm{adv}}$, and* $\sup_t |e(t)| \le \sup_t |e_{\mathrm{ben}}(t)| + \sup_t |e_{\mathrm{adv}}(t)|$.

Proof. Superposition for a linear time-varying system, then the triangle inequality. □

Two remarks bound the lemma's reach. First, actuator saturation is not linear, so the lemma holds only inside the region where (C1) holds; the parent paper excludes saturation from that region for the same reason [9]. Second, outages and loss of effectiveness are *parametric* sources: they change the system rather than add an input. The further decomposition of e_{ben} into per-source suprema, $\sup_t |e_{\mathrm{ben}}| \le \sum_i X_i$, is therefore an assumption, which we state as condition (C5) and test mission by mission in Section VII.

B. Dependence-sharp aggregation

Because the X_i share no known joint law, we bound the tail of their sum over *all* couplings consistent with the marginals. For non-negative $X_1,\dots,X_d$ with survival functions $S_i(t) = \Pr[X_i > t]$, define the worst-case and standard bounds

$$\overline{p}(s) = \sup_C \Pr_C\Big[\textstyle\sum_i X_i > s\Big], \tag{4}$$

$$p_U(s) = \min\Big\{1,\ \inf_{s_1+\cdots+s_d=s} \textstyle\sum_i S_i(s_i)\Big\}, \tag{5}$$

where the supremum runs over all copulas C.

Proposition 2 (Tightness of the union bound)**.** *(a) $\overline{p}(s) \le p_U(s)$ for all s. (b) For $d = 2$, $\overline{p} = p_U$ at every continuity point: the union bound is sharp [15], [16], [27]. (c) For $d \ge 3$ the inequality can be strict; the Rearrangement Algorithm (RA) approximates $\overline{p}$, with excellent empirical accuracy but no general convergence proof [28], [29]. (d) Against a known benign coupling C_0, the ratio $p_U(s)/\Pr_{C_0}[\sum X_i > s]$ never exceeds d under comonotonicity (take $s_i = F_i^{-1}(u)$) and equals d for identically distributed comonotone sources whenever the equal split is optimal. Under independence it is unbounded: for two independent* $\mathrm{Exp}(1)$ *sources, $p_U(s) \approx 2e^{-s/2}$ while the truth is $(1+s)e^{-s}$. (e) A deterministic term $X_j \equiv c$ has a single coupling with any other variable, so it shifts s by c and contributes no dependence gap.*

Proof. (a) For any split, $\{\sum X_i > s\} \subseteq \bigcup_i \{X_i > s_i\}$. (b) is Makarov's theorem [15], proved independently by Rüschendorf [16]; Frank et al. give the attaining coupling [27]. (c) Embrechts et al. give examples of strict inequality and the RA [29]. (d) is direct computation; for independent exponentials, p_U is attained at the equal split and the ratio $2e^{s/2}/(1+s)$ diverges. (e) The Fréchet class of a Dirac marginal is a singleton. □

a) Who chooses the coupling?: The adversarial offset enters as a deterministic term, so by (e) it has no coupling to choose. The dependence-sharp bound is needed for two other reasons. First, nature's coupling of gusts, outages and faults has never been measured. Second, an attacker who controls a *stochastic* source, for example by jamming to induce the outages and noise of δ_{gnss}, chooses when that source fires relative to the others, and hence its coupling, provided the induced episodes leave the marginal of X_{gnss} within its calibration (C3). Our proposal had hypothesised that a joint treatment recovers the union bound's slack precisely when the sources are comonotone. Proposition 2 corrects this. When the coupling is unknown or adversarial, the union bound is already the right quantity for two sources and nearly right for three. The slack exists only relative to a coupling that is *known* to be benign.

C. The floor

Theorem 1 (Source-agnostic, dependence-sharp bound)**.** *Assume (C1) a Lipschitz, unsaturated closed loop; (C2′) a trusted re-anchoring fix at least every T_a seconds, including at the end of the mission; (C3) transfer of the calibrated interface quantities (γ, the marginals of X_i, and the stealth budget δ_{adv}) to the deployment; (C4) an environmental marginal in the declared envelope, with CDFs $\underline{F}_{\mathrm{env}} \le F_{\mathrm{env}} \le \overline{F}_{\mathrm{env}}$; and (C5) sub-additivity across benign sources. Let U be the event that the stack Ψ_θ raises no alarm. Then, for every coupling of the benign sources and every attack,*

$$\Pr[\mathit{fail} \wedge U] \ \le\ \overline{p}\big(s_m\big), \qquad s_m = m - e^{LT_c}\delta_{\mathrm{net}}(\theta) - \delta_{\mathrm{adv}}(\theta), \tag{6}$$

with $\overline{p}$ evaluated at the stochastically largest environmental marginal, whose CDF is $\underline{F}_{\mathrm{env}}$. Consequently $\Pr[\mathit{complete} \mid U] \ge 1 - \overline{p}(s_m)/\Pr[U]$*, and in the population of all missions* $\Pr[\mathit{fail} \wedge U] \le 1 - f(m)$ *with* $f(m) = 1 - \overline{p}(s_m)$.

Proof. On U, (C3) gives $\sup_t |e_{\mathrm{adv}}| \le \delta_{\mathrm{adv}}$, so by Lemma 2 and (C5), failure on U implies $\sum_i X_i > s_m$. Hence $\Pr[\mathrm{fail} \wedge U] \le \Pr[\sum_i X_i > s_m] \le \overline{p}(s_m)$, which needs no conditioning on U. The tail of a sum is non-decreasing under first-order stochastic dominance of any summand, for every coupling and hence for the supremum, so the stochastically largest member of the p-box is the worst. The conditional form follows by dividing by $\Pr[U]$. □

Remark 1. The theorem is conditional in exactly the sense of the parent guarantee [9]: its first and last links are proved, and its interface quantities are measured. We therefore test (C3)–(C5) empirically in Section VII instead of asserting them. Bounding the *joint* event $\{\mathrm{fail} \wedge U\}$, rather than the completion rate conditional on silence, is what lets the bound use unconditional benign marginals: conditioning on silence would select benign realisations and change their law.

D. Computing the floor

The marginals are estimated on a calibration split (seed block 1xxx) from 4,000 missions per source. p_U is minimised over a grid of splits. $\overline{p}$ is approximated by the RA with $N = 1{,}000$ quantile levels, using both the upper and the lower discretisation, keeping the larger value, and bisecting on the confidence level [29], [30]. p_U is a proven upper bound on $\overline{p}$; the RA value, with the conservative discretisation, is a numerical approximation of $\overline{p}$ that is not proven in general. We report $\min(\mathrm{RA}, p_U)$ as our estimate of $\overline{p}$, which also removes the RA's $1/N$ resolution floor in the far tail, and we give p_U alongside wherever a bound must be strictly certified. For reference only, we also report the tail under independence (Monte Carlo) and under comonotonicity (quantile sum). Neither is a certificate.

V. The Cross-Layer Benchmark

A guarantee that spans layers needs data objects that record the coupling between them, and no public corpus provides such objects. The benchmark extends the parent's two-layer schema [9]. Its three record kinds are an *Event* (an atomic observation), an *AttackWindow* (a labelled interval) and a *CrossLayerPairing*, which joins a network or environmental window to the navigation window it induces. A pairing carries the detector operating point, the residual budget, the induced perturbation and the mission outcome. Version 0.2 adds a `perturbation_source` field taking values `net`, `gnss`,

`fault`, `env` or `adv`, so that fault and environment pairings share the attack pairings' representation.

a) Provenance is mandatory.: Every pairing carries one of three flags: `measured_same_platform` (both windows observed on one airframe), `physics_model` (coupled through a stated physical relation) or `synthetic_alignment` (aligned in simulation). Readers can therefore filter by strength of evidence rather than trust an unaudited join. The flag keeps the most important qualification in plain view. Of the six corpora, only the UAV Attack Dataset [3], [31] observes a network or radio event and the flight it degrades on the same platform, and its release carries no machine-readable attack intervals. The number of released `measured_same_platform` pairings is therefore zero, and every record of the benchmark says so.

b) Contents.: Lossless adapters ingest the HCRL UAVCAN bus corpus [2], UAVIDS-2025 [1], UAV-CAS [32], a replay of the DATAMUt time-delay detector [33], the UAV Attack Dataset, and the physics-informed UAV-EW-Bench [34]. Together these give 431,773 events, of which 200,610 were captured on real hardware. Each adapter is verified by a hash round-trip, so that no source column is discarded. Fault and environment sources enter through the RflyMAD [35] and ALFA [36] fault taxonomies and the Dryden model [20]. The benchmark ships a detector-operating-curve harness that sweeps θ, the dependence-bound library (union, RA, independence, comonotone), the disguise-evasion harness of Section VI, and datasheets [37] that map each pitfall of Arp et al. [38] to its mitigation. Third-party data is fetched from its origin under its own licence. We release adapters and unified labels, never redistributed corpora. In this paper the benchmark supplies the interface constants of the network term (γ, s, L; the event totals come from the parent's provenance ledger). The evasion study itself runs in simulation, and its records enter the benchmark with the provenance flag `synthetic_alignment`.

VI. Evasion Study Design

A. Closed loop and perturbation models

The pilot models the lateral (cross-track) channel of a small multirotor cruising at $V = 8\,\mathrm{m/s}$ through a 60 s mission segment. The airframe is a drag-damped double integrator, $\ddot{p} = u + c_d(w - \dot{p})$ with $c_d = 0.35\,\mathrm{s}^{-1}$. It is driven by a proportional–derivative (PD) controller acting on the KF estimate and saturated at $4\,\mathrm{m/s}^2$. The KF has states (position, velocity, accelerometer bias), propagates with the 20 Hz accelerometer, and corrects with 10 Hz GNSS fixes and the anchor fix every $T_a = 5\,\mathrm{s}$.

Environment. The lateral wind w follows the Dryden spectrum of MIL-F-8785C at 100 ft altitude. With W_{20} the wind speed at 20 ft, it has $\sigma_v = 0.1W_{20}/(0.177 + 0.000823h)^{0.4}$ and scale length $L_v = h/(0.177 + 0.000823h)^{1.2}$. The declared envelope is $W_{20} \in [15, 30]$ kt (light to moderate); severe turbulence (45–60 kt) lies outside it and serves as a negative control. Because the accelerometer measures the aerodynamic force, wind is consistent between the IMU and GNSS, as it is on a real airframe to first order.

Faults. Faults are drawn independently per mission: an accelerometer bias step (probability 0.3, $\sigma = 0.1\,\mathrm{m/s}^2$); bias drift (0.2, with the random walk inflated tenfold); and actuator loss of effectiveness (LoE) of 10–30% (0.2).

GNSS. Noise has a white component ($\sigma = 0.5\,\mathrm{m}$) and a first-order Gauss–Markov drift ($\sigma = 0.8\,\mathrm{m}$, $\tau = 60\,\mathrm{s}$). Outages are exponential with mean 5 s (probability 0.3), and jamming-noise episodes triple σ for 10 s (probability 0.2).

B. Detectors

All four detectors observe only $O_{0:T}$. Each reduces a mission to one statistic S_D and alarms when $S_D > \theta_D$:

D1, signature monitor.
$S_1 = \max_k |z_k - z_{k-1}|$, the largest jump between consecutive fixes. It stands in for rule-based IDSs tuned to malicious signatures such as jumps and discontinuities.

D2, χ^2 innovation gate.
This is loosely modelled on innovation gating in PX4-class filters: it alarms when the normalised innovation squared (NIS) ν_k^2/S_k exceeds θ_2 on three consecutive fixes. (PX4's EKF2 rejects individual measurements rather than alarming.)

D3, SAVIOR-style CUSUM.
A model-based observer driven by the commanded acceleration with a no-wind model accumulates a cumulative-sum (CUSUM) statistic $g_k = \max(0, g_{k-1} + r_k^2 - (1+\kappa))$ on its GNSS residual [5].

D4, IMU–GNSS invariant.
A quadratic is fitted over 4 s to the residual between the GNSS position and the doubly integrated accelerometer; the detector alarms when the curvature, an acceleration discrepancy, exceeds θ_4 on five consecutive fixes. It is the cross-sensor analogue of the control invariants of Choi et al. [4].

Whenever C2′ holds, an anchor gate on the anchor fix's NIS (calibrated FPR 1%) is part of every stack. Thresholds are set on 4,000 benign missions so that each detector has the stated per-mission FPR; the combined stack ALL splits the budget equally among its four members.

C. Disguise families and the attacker

The offset $d(t)$, applied from an onset time t_0, belongs to one of five families:

- *step*: $d = A$ (the naive spoofer);
- *ramp*: $d = \min\{r(t - t_0), A\}$ (the "boiling frog");
- *fault mimic*: $\ddot{d} = \beta_0 + \beta(t)$, with β a random walk, capped at $|d| \le A$. It imitates accelerometer-bias drift, which the filter's bias state absorbs;
- *GNSS mimic*: $d = A|g(t)| + c(t - t_0)$, with g a unit Gauss–Markov process of attacker-chosen correlation time, imitating multipath or ionospheric drift;
- *weather mimic*: $d = A\,w(t - \ell)/\sigma_{\mathrm{mod}} + b\min\{(t - t_0)/10, 1\}$, where σ_{mod} is the moderate-turbulence σ_v. The offset follows the true gust with a lag ℓ.

The attacker solves (3) with the cross-entropy method (CEM) [39]: 12 candidates per generation, 4 elites, 25 gener-

ations and 48 missions per candidate. Each (stack $\times$ family $\times$ operating point) combination gets its own optimiser seed in block 2xxx; stronger optimisers (CMA-ES [40], Bayesian optimisation) are runbook item R1. Because a local optimiser can fail to find an evading parameterisation that exists, we do not read detector strength off the optimiser's own target. Instead, we *cross-evaluate* every optimised attacker against every stack.

D. Metrics and statistical protocol

For each optimised attacker, evaluated on fresh missions (block 4xxx), we report:

- the *evasion rate* (no alarm during the whole mission);
- the attack's own contribution $\sup_t |e_{\text{adv}}|$ on evading missions, computed on the same realisation with the additive benign inputs set to zero but the parametric ones (outages, LoE) kept, as in Lemma 2;
- the Kolmogorov–Smirnov (KS) distance between each detector's statistic under attack and under benign conditions, a finite-sample estimate of the TV distance of Lemma 1 restricted to threshold tests.

In the cross-evaluated disguise matrix, a cell's contribution is the maximum over attackers that evade in at least 5% (and at least 10) of missions; we report the unfiltered maxima separately. The stealth budget $\delta_{\text{adv}}(\theta)$ is the supremum of the attack-only deviation over *all* evading missions, with no such filter. It is pooled over all 35 optimised attackers on calibration missions (block 1xxx). Every attacker uses the same missions at every operating point, and an attack that is stealthy against stricter thresholds is also stealthy against looser ones, so the pooled supremum is monotone in the operating point by construction.

For soundness, we re-optimise an independent attacker with fresh seeds (block 3xxx) for every family and FPR, against the full stack. We evaluate it on $n = 1{,}000$ fresh missions per operating point, crossing family, FPR, wind (light, moderate, envelope), margin m and θ_{net}. The theorem's event is $\{\text{fail} \wedge U\}$, so we count its occurrences k and compute the one-sided binomial p-value of H_0: $\Pr[\text{fail} \wedge U] \le \overline{p}(s_m)$. The p-values are adjusted across all operating points by Holm's procedure [41], and a violation is a Holm-adjusted $p < 0.05$. We also report the MCR conditional on U, with Clopper–Pearson intervals [42].

VII. Evaluation

The pilot comprises 1,909,600 simulated missions and ran in 29 minutes on two CPU cores. Every number below is generated from the released result files.

A. Calibration: who sets the thresholds?

At the 5% operating point, the per-detector FPR stays between 3.8% and 6.9% under moderate wind, and the anchor gate alone contributes 0.95%. One finding shapes everything that follows: *wind and faults alone trip no detector* (FPR 0.0% for every detector when either is the only benign source), whereas GNSS error alone produces up to 5.3%. The thresholds are therefore set by the tails of the GNSS error, and the tolerance they grant is the space an attacker exploits. Severe turbulence, outside the declared envelope, raises the FPR of the SAVIOR-style detector D3, which models no wind, to 8.2%; the others stay at or below 6.0%.

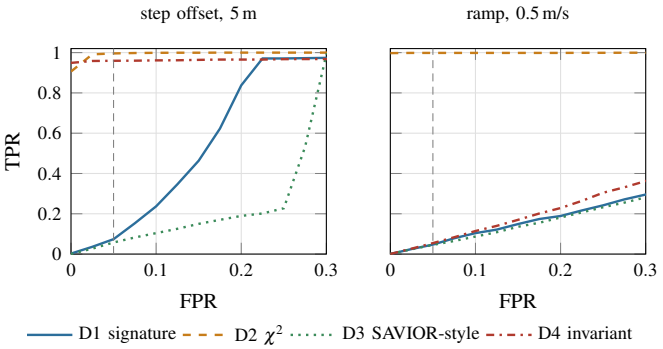


Fig. 2. Detector operating curves against two *naive* offsets, with thresholds swept over 4,000 benign missions (dashed line: the 5% operating point). Only D2 and D4 separate a naive 5 m step, and only D2 separates the ramp.

Figure 2 shows the operating curves against naive offsets. At 5% FPR the χ^2 gate detects a 5 m step in 100% of missions and the invariant D4 in 96%, but the signature monitor and D3 detect it in only 7% and 6%, which is close to chance. A 0.5 m/s ramp is caught by D2 (100%) and essentially by no other detector ($\le$5%).

B. RQ1: evasion is universal; the damage is not

a) Cross-evaluation.: We evaluated each of the 35 optimised attackers against every stack. In 8 of 25 (stack, family) cells, the CEM run that targeted that stack found no parameterisation evading in at least 5% of missions, although an attacker optimised for a different stack does. Reading detector strength off the optimiser's own target would therefore have been wrong, so Table I reports the best attacker of each family, wherever it was optimised.

b) With the anchor gate, the anchor sets the budget.: With condition C2′ in force and its gate armed (Table I, top), every one of the five families evades every detector, and the full stack, in at least 87% of missions; for reference, benign missions raise no alarm in about 94% under the full stack. Each family has parameterisations that offset GNSS by about 1–2 m, and every detector must tolerate offsets of that size to keep its FPR at 5%. Among attackers that evade in at least 5% of missions, the largest undetected contribution lies between 2.1 and 2.9 m *whichever detector is deployed*. Rarer evasions go further: up to 5.6 m against D1, D3 (weather mimic). The pooled supremum against the full stack is 4.4 m (Fig. 3), and the stealth budget is $\delta_{\text{adv}} = 4.37$ m. Within that range the trusted fix, not the detector design, decides how far an undetected attacker gets. The median KS distance between a detector's statistic under an evading attack and under benign conditions is 0.07, against 0.99 for detected ones. This is the empirical face of Lemma 1: an attack that evades is, to the detector, almost indistinguishable from the benign population.

TABLE I

DISGUISE MATRIX AT 5% FPR, CROSS-EVALUATED OVER ALL 35 OPTIMISED ATTACKERS ON 1,000 HELD-OUT MISSIONS PER ATTACKER. *Top:* ANCHOR GATE ARMED; BEST EVASION RATE (%) / LARGEST UNDETECTED ATTACK CONTRIBUTION (M). *Bottom:* ANCHOR FUSED BUT ITS GATE REMOVED; LARGEST UNDETECTED CONTRIBUTION (M). CONTRIBUTIONS ARE TAKEN OVER ATTACKERS THAT EVADE IN AT LEAST 5% (AND AT LEAST 10) OF MISSIONS; RARER EVASIONS ARE REPORTED IN THE TEXT. BENIGN MISSIONS RAISE NO ALARM IN ABOUT 94%. DARKER CELLS ARE EVADED MORE OFTEN (TOP) OR ADMIT MORE DEVIATION (BOTTOM).

	step	**ramp**	**fault mimic**	**GNSS mimic**	**weather mimic**
anchor gate armed (C2′)					
D1 signature	89 / 2.1	91 / 2.4	94 / 2.6	92 / 2.5	88 / 2.9
D2 χ^2 gate	88 / 2.1	90 / 2.4	93 / 2.6	91 / 2.5	88 / 2.9
D3 SAVIOR-style	89 / 2.1	89 / 2.4	94 / 2.6	90 / 2.5	88 / 2.8
D4 IMU–GNSS inv.	87 / 2.1	90 / 2.4	92 / 2.6	91 / 2.5	89 / 2.9
ALL (stack)	88 / 2.1	90 / 2.4	94 / 2.6	91 / 2.5	90 / 2.9
anchor fused, gate removed					
D1 signature	2.2	2.4	73.9	72.6	50.4
D2 χ^2 gate	2.2	2.4	9.9	2.6	2.9
D3 SAVIOR-style	2.2	2.4	73.9	37.9	37.2
D4 IMU–GNSS inv.	2.2	2.4	73.9	2.6	3.3
ALL (stack)	2.2	2.4	10.5	2.6	2.9

c) Without the gate, each detector admits a different disguise.: The bottom of Table I repeats the cross-evaluation with the anchor still fused into the filter but its alarm removed, isolating what the detectors themselves discriminate. Evasion stays at or above 94%, but the undetected contribution now depends sharply on the pairing of disguise and detector.

- The fault mimic drives 73.9 m past D1, D3 and D4. Its slow, bias-shaped curvature has no jumps for D1, matches the bias drift that D4 tolerates by design, and looks like an unmodelled disturbance to D3.
- The GNSS mimic and the weather mimic exploit the two detectors that do not cross-check the IMU, D1 and D3 (up to 72.6 and 50.4 m). Among frequently evading attackers, D4 caps them at 3.3 m or below, because their acceleration content contradicts the accelerometer; a rare GNSS-mimic evasion still reached 19.2 m.
- Rare step offsets that happen to land in a benign noise trough evade D1, D3 and D4 in about 3% of missions and reach 26.8 m.
- Only the χ^2 gate D2, whose filter models the accelerometer bias, caps every family, at 9.9 m (fault mimic), including rare evasions (9.9 m); the full stack, which contains D2, is capped at 10.5 m.

This is the pattern adversarial equivalence predicts. The disguise that wins is the one the detector models as benign: bias drift for an invariant that tolerates bias, and drift or wind-shaped motion for detectors without an inertial cross-check.

d) The terminal window.: Figure 3 sweeps the re-anchoring interval against the full stack, pooling all attackers on common missions. With every interval closed, the largest undetected contribution grows from 4.4 m at $T_a = 5$ s to 20.6 m at 20 s (4.7×). The attacker must return near zero before each trusted fix, so the budget is set by how far it can travel and come back within one interval. When the segment before mission end is left unchecked, no return leg is needed, and the contribution reaches 137.8 m, 7× the closed value. The terminal window is where an equivalence attacker spends its budget.

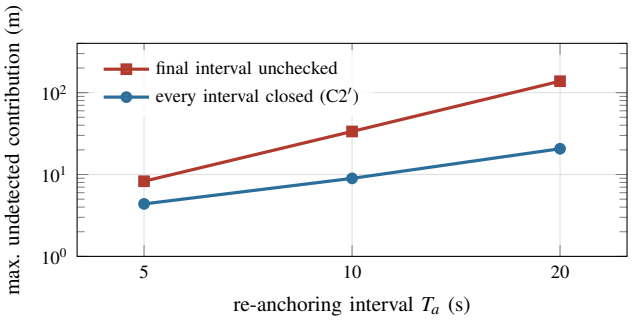


Fig. 3. Stealth budget against the full stack (5% FPR) versus the re-anchoring interval, pooled over 30 attackers (15 tuned per T_a and the 15 optimised against the full stack) and maximised over the five disguise families, on the same missions at every T_a. With every interval closed, the largest undetected contribution grows from 4.4 to 20.6 m (4.7×). With the final segment before mission end unchecked, it grows 17× to 137.8 m.

C. RQ3: where the looseness lives

Figure 4 decomposes the tail of the benign budget; the ratios below are at $s = 4$ m, and their ordering changes in the far tail, where few joint exceedances remain.

1) For two sources, the union and RA bounds agree to within 0.001, as Makarov's theorem requires. For three, the RA tightens the union bound by 20% (0.6390 to 0.5133). Under unknown dependence, the union bound is nearly the right quantity.
2) The worst coupling exceeds independent sums at the same marginals by 4× (*dependence gap*). Evaluating the wind at the upper envelope rather than at the conditions actually flown costs a further 2.2× (*envelope gap*).
3) Independent sums at matched wind agree with the sum of per-source suprema measured on the same missions to within 1.09×, consistent with the benign sources being independent, as they are by construction in the simulator.
4) The sum of per-source suprema exceeds the supremum of the joint deviation by 16.6× (*temporal gap*; 62 joint exceedances in 20,000 missions). Sources seldom peak at the same instant, but a budget built from per-source suprema must assume they do.

The dependence gap can be closed only by *knowing* the coupling, which an attacker who controls jamming denies the defender. The envelope gap shrinks with better wind knowledge. The temporal gap, the largest of the three at this s, can be closed without either, by bounding the supremum of the summed trajectory directly, for example with a reachable-set computation per certification window. Condition (C5), which licenses adding per-source suprema, held in 99.63% of joint missions.

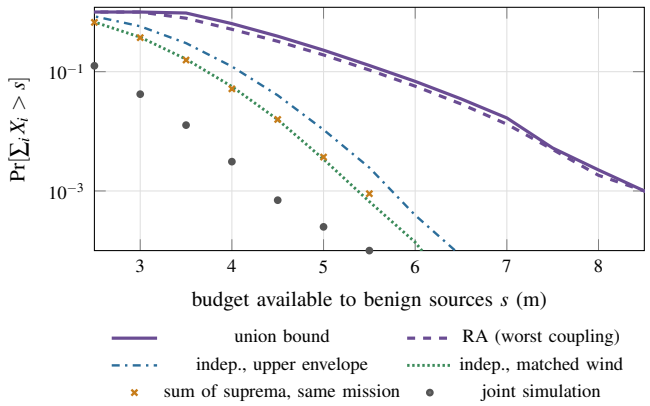


Fig. 4. Where the looseness lives (tail of the benign budget). Union and RA nearly coincide. The steps down from RA are: to independent sums at the upper wind envelope (dependence gap); to independent sums at the wind actually flown (envelope gap); to the sum of per-source suprema on the same missions, which confirms that the simulated sources are independent; and to the joint simulation (temporal gap: sources seldom peak at the same instant).

D. RQ2 and RQ4: the bound under evasion

Pooling all attackers on calibration missions gives a stealth budget of $\delta_{\text{adv}} = 4.37$ m, attained by the fault mimic. The budget is identical at 1%, 5% and 10% FPR. It is a single pooled maximum over common missions, so this says that no attacker in the pool gains from looser thresholds in that range, not that thresholds never matter.

Table II reports, for each operating point, the bounds on $\Pr[\text{fail} \wedge U]$ and the largest observed frequency of that event over five held-out attackers and three wind conditions. Across 270 distinct operating points and 30,541 undetected-attack missions, **no point shows a Holm-significant violation** (0 of 270). The scaled test of Section VII-E overturns this. Wherever the bound is informative ($\overline{p} < 1$), the observed frequency of undetected failure was zero. The bound is therefore conservative, and the zero violations by themselves carry little evidence: the negative controls below are what show that the test can fail.

Three further checks belong with the result.

- *The empirical budget is not a theorem.* The held-out attacker exceeded δ_{adv} in 1 of 45 (family × FPR × wind) cells, reaching 5.13 m with the fault mimic at 5% FPR (1.17×). The bound absorbed it, but this is why we strengthened the optimisers and derived an analytic budget (Sections VII-E and VII-F). In 6 cells, the held-out optimiser found no evading attack at all.
- *Lemma 2 holds where (C1) holds.* Attack/benign superposition held in at least 94.4% of missions at every point. A control run with the actuator limit removed (5% FPR, envelope wind, all five families) restored it in 100.00% of missions (largest excess 7.8e-15 m). We attribute the larger excesses at other points, up to 11.03 m, to the same mechanism, saturation, which (C1) excludes; they were not re-run.
- *Independence is not certified.* The independence bound $p_\perp$ was not violated either, because the simulated sources are independent. It would claim a 0.9 floor at $m = 8.5$ m, against 10.0 m for the certified bound (Fig. 5); nothing guarantees that independence when an attacker controls jamming.

TABLE II

SOUNDNESS OF THEOREM 1 ON HELD-OUT ATTACKERS. BOUNDS ON $\Pr[\text{FAIL} \wedge U]$: UNION p_U, RA $\overline{p}$ (CERTIFIED) AND INDEPENDENCE $p_\perp$ (NOT CERTIFIED). OBS. = LARGEST OBSERVED FREQUENCY OF $\{\text{FAIL} \wedge U\}$ OVER FIVE FAMILIES AND THREE WIND CONDITIONS; MCR_U = SMALLEST COMPLETION RATE AMONG UNDETECTED MISSIONS; VIOL. = HOLM-SIGNIFICANT VIOLATIONS OF $\overline{p}$ AND OF $p_\perp$. $\delta_{\text{adv}} = 4.37$ M. ROWS WITH $\theta_{\text{net}} = 0.25$ S SHIFT THE MARGIN ANALYTICALLY BY $e^{LT_c}\delta_{\text{net}} = 2.65$ M; THE NETWORK TERM ITSELF IS NOT SIMULATED. p_U IS A PROVEN UPPER BOUND; THE RA COLUMN IS A NUMERICAL APPROXIMATION OF THE SHARP $\overline{p}$, REPORTED AS $\min(\text{RA}, p_U)$.

FPR	θ_{net}	m	p_U	$\overline{p}$	$p_\perp$	obs.	MCR_U	viol.	
1%	0.00	5	1.000	1.000	1.0000	0.0030	0.996	0	0
1%	0.00	10	0.109	0.091	0.0016	0.0000	1.000	0	0
1%	0.00	20	0.000	0.000	0.0000	0.0000	1.000	0	0
1%	0.25	5	1.000	1.000	1.0000	0.4640	0.432	0	0
1%	0.25	10	1.000	1.000	0.5878	0.0000	1.000	0	0
1%	0.25	20	0.000	0.000	0.0000	0.0000	1.000	0	0
5%	0.00	5	1.000	1.000	1.0000	0.0020	0.998	0	0
5%	0.00	10	0.109	0.091	0.0016	0.0000	1.000	0	0
5%	0.00	20	0.000	0.000	0.0000	0.0000	1.000	0	0
5%	0.25	5	1.000	1.000	1.0000	0.4250	0.508	0	0
5%	0.25	10	1.000	1.000	0.5878	0.0000	1.000	0	0
5%	0.25	20	0.000	0.000	0.0000	0.0000	1.000	0	0
10%	0.00	5	1.000	1.000	1.0000	0.0010	0.999	0	0
10%	0.00	10	0.109	0.091	0.0016	0.0000	1.000	0	0
10%	0.00	20	0.000	0.000	0.0000	0.0000	1.000	0	0
10%	0.25	5	1.000	1.000	1.0000	0.3880	0.467	0	0
10%	0.25	10	1.000	1.000	0.5878	0.0000	1.000	0	0
10%	0.25	20	0.000	0.000	0.0000	0.0000	1.000	0	0

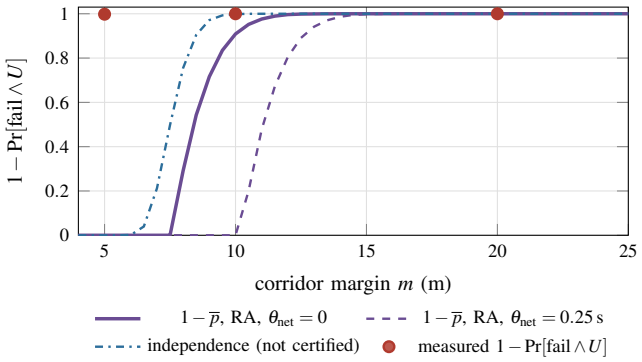


Fig. 5. Certified bound versus measurement at 5% FPR. The measured points are the worst case over five held-out attackers and three wind conditions. The bound stays below every measurement. Including the network term shifts it right by $e^{LT_c}\delta_{\text{net}} = 2.65$ m.

a) Negative controls: does the test have power?: A soundness test that cannot fail is uninformative, so we broke two conditions deliberately and re-optimised the attacker under each (Table III).

- *(C2′) broken, no re-anchoring.* Undetected deviation reaches 308.5 m, and the test flags 4 of 6 points (fault and ramp), every one where the bound is informative.
- *(C4) broken, severe turbulence (45–60 kt).* The fault mimic reaches 96.7 m undetected, and the test flags the bound at

TABLE III
NEGATIVE CONTROLS AT 5% FPR. UNDET. = UNDETECTED MISSIONS OUT OF 1,000; FAILS = FAILURES AMONG THEM; VIOL. = HOLM-SIGNIFICANT VIOLATION OF $\overline{p}$; DEV. = LARGEST UNDETECTED DEVIATION (M).

Control	disguise	m	undet.	fails	$\overline{p}$	viol.	dev.
C2′: no anchor	ramp	5	907	907	1.000	no	47.8
C2′: no anchor	ramp	10	907	907	0.091	**yes**	47.8
C2′: no anchor	ramp	20	907	907	0.000	**yes**	47.8
C2′: no anchor	fault	5	729	719	1.000	no	308.5
C2′: no anchor	fault	10	729	683	0.091	**yes**	308.5
C2′: no anchor	fault	20	729	361	0.000	**yes**	308.5
C4: severe wind	fault	5	889	106	1.000	no	96.7
C4: severe wind	fault	10	889	37	0.091	no	96.7
C4: severe wind	fault	20	889	14	0.000	**yes**	96.7
C4: severe wind	weather	5	0	0	1.000	no	0.0
C4: severe wind	weather	10	0	0	0.091	no	0.0
C4: severe wind	weather	20	0	0	0.000	no	0.0

$m = 20$ m, where $\overline{p} = 0$ because the empirical marginals have finite support. At $m = 10$ m the observed frequency stays below $\overline{p}$, so this break is detected only where the bound is tightest. The weather mimic was detected in every mission, so there was nothing to test.

The test thus detects unsoundness when a stated condition fails and enough undetected missions remain to measure it, which is the behaviour the audit loop of Fig. 1 relies on.

E. RQ5: the test at scale breaks the pilot instantiation

The pilot's zero violations rested on five CEM attackers per cell and a conservative bound. We therefore strengthened the attacker and the test before drawing conclusions (runbook R1). 126 further attackers were optimised against the full stack: CMA-ES (population 16, 200 generations, one restart) [40], Bayesian optimisation with a tree-structured Parzen estimator (400 trials), and the same TPE with a tail objective (the 90th percentile of the undetected deviation instead of its mean). Every candidate was scored on 128 missions with common random numbers, and a sixth *mixed* family switches between any two disguises mid-mission. The scaled soundness test re-optimised 5 independent CEM attackers per (family, FPR) cell and evaluated each on 5,000 fresh missions under three wind conditions: 1,620 tests over 797,435 undetected-attack missions.

a) The stealth budget grew by 25%.: The pooled budget rose from 4.37 to 5.46 m at 1% and 5% FPR (1.25×) and to 5.04 m at 10%, attained by the gnss mimic (tail-objective TPE). Only the tail objective found more than the pilot's CEM: the mean-objective CMA-ES and TPE peaked at 3.85 and 3.09 m. An attacker that optimises its typical deviation is not the attacker that sets an empirical supremum. Even the larger budget was exceeded by a held-out attacker in 12 of 270 cells, by the mixed family up to 7.04 m. An empirical supremum is a lower bound on δ_{adv}, and more search keeps raising it.

b) Five Holm-significant violations.: With $\delta_{\text{adv}} = 5.46$ m, the test flagged 5 of 1,620 points, all at $m = 20$ m, where the bound claimed $\overline{p} = 0$. We re-simulated each flagged cell with its seeds (`r1_violation_diagnosis.csv`). The violations come from 3 missions, and the attacker caused none of them. Each fails under its benign inputs alone (benign-only deviation 19.2–34.5 m), the attack adds at most 2.3 m, and all three saturate the actuator with or without the attack. Two conditions of Theorem 1 failed, and neither is about disguise.

- *(C3) for the marginals.* The empirical marginal of each source has finite support. Beyond the largest of 4,000 calibration samples, $\overline{p}$ is exactly zero, and the bound claims certainty that no sample can justify. One benign mission in 5,000 fails at $m = 20$ m.
- *(C1), which nothing enforced.* Superposition held in at least 90.9% of missions, and every violating mission saturated. The pilot had stated (C1) as an assumption, but no component of the stack checked it at run time.

This is a negative result about the pilot instantiation, not about Theorem 1. The theorem's conclusion requires its conditions, and the scaled test found two that the instantiation did not secure. We report the violation as found. The revision below fixes the instantiation before looking at new data and does not tune the old run until it passes.

F. A revised instantiation, judged on fresh data

We changed three things, fixed them before generating any new data, and evaluated them on a new seed block. A smoke test of the code used a separate block, which we did not reuse.

1) *A saturation alarm D_{sat}.* The autopilot raises an alarm at any epoch at which the commanded acceleration reaches its limit. This turns (C1) from an assumption into a run-time check: on the event U, the loop never saturated, so Lemma 2 applies exactly. The alarm costs benign missions 1.1% in false alarms (5% operating point, envelope wind). The full stack's false-alarm rate rises from 4.8% to 5.6%.
2) *A finite-sample p-box for (C3).* Each marginal tail is replaced by its Clopper–Pearson upper bound, taken simultaneously over the split grid (Bonferroni over 3×501 grid points, $\alpha = 0.05$), and aggregated by the union bound. With probability at least 0.95 over the calibration samples, $\overline{p}$ is then a valid upper bound everywhere on the grid. It never reaches zero, and with 4,000 samples per source it levels off at $1 - 0.992$.
3) *Two stealth budgets, tested separately.* The first is the R1 calibration supremum, $\delta_{\text{adv}} = 5.46$ m. It was *not* widened to the 7.04 m that the R1 held-out attacker reached, because a constant widened after looking at held-out data would make the new test circular. The second is the analytic bound below, which holds for every attack.

a) An analytic stealth budget (R8).: With noise removed, the linear loop maps the attack offsets d_j at the GNSS epochs to the attack-only deviation linearly. The worst undetected deviation is therefore a linear programme over d, solved once for every time step. Its constraints are the detector conditions that are convex in d and *implied* by silence, with the benign contribution bounded by its $1 - \varepsilon/3$ calibration quantile ($\varepsilon = 0.01$).

- D1: a bounded jump between consecutive fixes.
- The anchor gate: a bounded innovation at each trusted fix.

- D_{sat}: a bounded attack-only command.

D2–D4 alarm on runs of exceedances or on quadratic statistics, which are not convex in d, so we drop them. Dropping a detector only enlarges the feasible set, so the bound stays valid for every stack that contains the kept ones. We take the worst actuator effectiveness in $[0.7, 1]$, and the onset of the attack is free. Three findings follow (`r8_analytic_bound.csv`).

- *The anchor gate alone bounds nothing.* The trusted fix constrains the estimate error at anchor times, not the deviation. An attacker that carries the estimate along with the vehicle passes every anchor.
- *A jump detector adds little.* With D1 the budget is finite but 303 m at $T_a = 5$ s, because D1's threshold must tolerate the jumps of GNSS jamming noise.
- *The saturation alarm is what makes it physical.* With D_{sat} the budget falls to 31.9 m. That is still 5.8× the calibrated δ_{adv} and 4.5× the largest deviation any optimised attacker reached (7.04 m).

The gap is the measure of what the non-convex detectors D2–D4 contribute and what no one has yet proved. Using the provable budget, the floor certifies nothing below a margin of about 31.9 m. Using the empirical budget, it certifies more, but rests on a supremum that more search has raised before.

b) Result on fresh seeds.: We re-optimised 72 held-out attackers against the revised stack: three CEM seeds per (family, FPR) cell plus one tail-objective TPE. Each was evaluated on 5,000 missions under three wind conditions, from a seed block never used before (510,848 undetected-attack missions).

- *No violation under either budget.* With the R1 budget, the test found 0 Holm-significant violations in 1,296 tests; with the LP budget, 0 in 1,296. Superposition was exact on every undetected mission (largest excess 0.0e+00 m). Missions like the benign-tail failures that broke the pilot instantiation saturate, so they now raise the alarm and leave the event U.
- *The empirical budget was exceeded again.* In 6 of 216 cells a held-out attacker went beyond $\delta_{\text{adv}} = 5.46$ m, up to 6.25 m (weather mimic, CEM). Benign slack absorbed it and no violation followed, but the stealth budget we can use is still not a sound constant. The LP budget was never exceeded. It can be exceeded only with probability ε, or on missions with a GNSS outage, which its proof does not cover.
- *What the bound certifies.* With the R1 budget, the floor at 5% FPR is 0.541 at $m = 10$ m and 0.992 at $m = 20$ m. With the LP budget it is 0.000 at both, so it certifies nothing in the corridors we test.
- *Limited power.* Of the 648 tests at which the bound is informative, 0 saw any undetected failure. The revision is therefore *not refuted*, not *confirmed tight*. The negative controls remain the evidence that the test can fail, and the scaled run of Section VII-E shows that it does when a condition is missing.

G. RQ6: how much does a measured GNSS marginal cost?

The pilot's GNSS marginal is simulated (99th percentile 2.8 m). The parent artifact contains measured navigation error during real GPS jamming: 496 one-second samples over two episodes, a noise episode and a loss of fix, whose error envelope grows with staleness to 6.2 m at 10 s and 7.0 m at 15 s (runbook R5; spoofing samples are excluded, since spoofing belongs to δ_{adv}). We replaced each simulated degradation episode by the larger of its simulated deviation and the measured envelope at the episode's staleness, with the simulator's episode probabilities, and in a stress variant set every episode to the measured 45 s duration. The construction is conservative in one respect, since it never makes a mission's GNSS deviation smaller. It also assumes that the measured navigation error becomes cross-track deviation one for one, which holds when the controller tracks its estimate and is not itself a measurement.

At $m = 10$ m and 5% FPR, with $\delta_{\text{adv}} = 5.46$ m, the floor estimated from the empirical marginals falls from 0.69 (simulated GNSS) to 0.34 (measured envelope) and 0.32 (45 s episodes). With the finite-sample band of Section VII-F, the values are 0.54, 0.21 and 0.19. The bound is only as good as its GNSS marginal. A simulated marginal overstated the certified floor by a factor of two to three at this margin. The jamming-induced tail, not the attack, is what the corridor must absorb. At $m = 20$ m every variant certifies 0.992, the ceiling that 4,000 calibration samples per source permit.

H. Summary

RQ1: every disguise family evades every detector; what differs is the damage. With the anchor gate armed, frequent evasions stay within 2.9 m and the pooled supremum is 4.4 m, whatever the detector. Without the gate, the damage depends on whether the detector models the disguise, from 9.9 m (D2) to 73.9 m (D1, D3, D4 against the fault mimic). **RQ2 and RQ4:** in the pilot, the bound held at all 270 held-out points, and the negative controls showed that the test can fail. **RQ3:** under unknown dependence the union bound is nearly sharp; the slack lies in the dependence, envelope and, above all, temporal gaps. **RQ5:** at scale, the pilot instantiation was unsound (5 violations). The cause was not disguise but two conditions it assumed without securing: a finite-sample marginal and an unchecked actuator limit. Once both were secured, the bound survived 72 fresh attackers with 0 violations, although the empirical δ_{adv} was exceeded again. The provable budget is 5.8× larger. **RQ6:** measured jamming data halves the certified floor at $m = 10$ m: a simulated GNSS marginal overstates the floor by a factor of two.

VIII. Discussion

a) What a defender should take away.: Five design rules follow from the measurements, and each is a mitigation of the attack the paper describes.

1) *Re-anchor, and size the interval rather than the threshold.* With a trusted fix every 5 s and its gate armed, frequent evasions contributed at most 2.9 m and the pooled supremum was 4.4 m, for every detector and disguise. Without the gate, the contribution reached 73.9 m. Lengthening the interval from 5 to 20 s multiplied the budget by 4.7 (Fig. 3), while moving the operating point from 1% to 10% FPR left the

pooled budget unchanged (Section VII). For a corridor of margin m, choose T_a such that $\delta_{\text{adv}}(T_a) \le m - e^{LT_c}\delta_{\text{net}} - s^\star$, where $s^\star$ is the smallest budget with $\overline{p}(s^\star)$ below the tolerated risk.

2) *Close the last interval.* An unchecked segment before mission end lets the attacker skip the return leg and multiplied the budget by 7 at $T_a = 20$ s. Landing and terminal approach are where a trusted fix matters most.
3) *Model the inertial bias, not just cross-check the inertial sensor.* Without the anchor gate, the only detector that capped every disguise was the one whose filter models the accelerometer bias (D2). D1 and D3 admitted tens of metres to the GNSS and weather mimics, and even the inertial invariant D4 admitted 73.9 m to the fault mimic, whose bias-shaped curvature it tolerates by design.
4) *Certify the undetected outcome, not the alarm.* Theorem 1 bounds $\Pr[\text{fail} \wedge U]$ for every attacker whose undetected contribution stays within δ_{adv}, without relying on the detector catching any particular disguise. A floor computed from benign marginals alone and gated on silence has no such property (Proposition 1).
5) *Alarm on what the certificate assumes.* The pilot assumed an unsaturated loop, and the scaled test failed exactly where the loop saturated. A saturation alarm turns the assumption into a run-time check, at a cost of 1.1% benign false alarms. Every condition of a deployed certificate should be either proved or monitored.

b) On the separability counter-claim.: Li et al. [43] and Xue et al. [44] argue that faults and attacks act through different subspaces and can be separated by residual design. Our gate-free results are compatible with that view for a *given* residual: D2 and D4 do separate several disguises. They differ in the quantifier. Separability holds for a fixed attack against a fixed detector. Equivalence concerns the attacker's best response, and against every detector we built, some family of small-amplitude offsets evaded in at least 87% of missions. Whether a residual exists that tolerates the benign classes and still rejects all of their imitations is open. Our evidence says only that none of ours does.

c) Limitations.:

- *Reduced-order model.* The pilot simulates one lateral channel with a linear KF. PX4's EKF2 has more states, different gating and nonlinear coupling. The disguise matrix and soundness must be re-established in SITL (runbook R3) before they are claimed for a deployed autopilot.
- *Empirical adversarial term.* The usable δ_{adv} is a pooled supremum over optimised attackers. Stronger optimisers raised it by 1.25×, and a held-out mixed-family attacker then exceeded it, reaching 7.04 m. The provable LP budget (Section VII-F) is 5.8× larger because it drops every non-convex detector. Tightening it requires reachable-set bounds for detectors that alarm on runs of exceedances, for example by mixed-integer programming, or detectors redesigned so that silence implies a convex constraint. We have done neither. A white-box gradient attacker through a differentiable simulator (runbook R1, step 3) was not run.
- *Finite calibration.* With 4,000 samples per source, the simultaneous band certifies at most 0.992. Certifying a floor of $1 - 10^{-k}$ needs several times 10^{k+1} calibration missions per source, which only simulation or fleet logs can supply.
- *Limited power.* At every point where the revised bound is informative, the observed frequency of undetected failure was zero. The test has shown that it can fail (the negative controls and Section VII-E), but the revised bound has not been stressed near its value. Closing the temporal gap would make it tight enough to be stressed directly.
- *GNSS marginal.* Replacing the simulated GNSS marginal by the measured jamming envelope (Section VII-G) halved the floor at $m = 10$ m. The fault marginal is still simulated: the RflyMAD and ALFA adapters of runbook R5 were not run, and the measured envelope comes from one campaign on one airframe.
- *Network term not simulated.* δ_{net} enters as the parent's certified tube; R4 injects measured relay staleness into SITL.
- *Covert attackers only.* The bound covers undetected failure. An attacker indifferent to detection is bounded instead by detection latency and the failsafe, which we do not certify here.
- *No IMU injection.* An attacker who can also inject into the IMU (acoustically [24], [25] or through bus reconfiguration [45]) defeats the inertial cross-checks, and condition C2′ then carries the whole burden.

IX. Related Work

a) Stealthy and disguised attacks on control systems.: The literature on stealthy attacks follows a line from false-data injection against χ^2-monitored Kalman filters [46], [47] and the attack space of Teixeira et al. [48], through impact limited by a false-alarm budget [49] and reachable-set security metrics [50], [51], to ε-stealthiness in KL divergence [13] and attacks that alarm at the false-alarm rate [14]. Pasqualetti et al. characterised undetectable attacks as unknown inputs [52]. Interval observers bound worst-case stealthy attacks under bounded noise [53], and secure state estimation bounds error under sparse attacks plus Gaussian noise [54]. Each treats the benign perturbation as a single, known nuisance. Roy and Dey formalised fault-mimicking actuator attacks in linear systems [10]. A recent dual-detection framework [44] and a unified secure-and-safe analysis [43] argue the opposite of our thesis: that faults and attacks occupy separable residual subspaces. Our cross-evaluation (Section VII) finds separability for particular detector–disguise pairs, but an attacker who chooses the disguise found an evading parameterisation against every detector we built. In power grids, discriminating disturbances from attacks [55], [56] and the observation that physical measurements alone cannot separate faults from attacks [57] parallel our setting. Load-altering attacks have been designed to stay robust to grid uncertainty [58], but not to hide inside it.

b) Attacks on UAV navigation.: GNSS spoofing has advanced from portable spoofers and takeover conditions [21],

[22] to covert UAV capture [23], adaptive hijacking [59] and attacks on multi-sensor fusion [60], [61]; see [62], [63] for surveys. Stealthy attacks against control-based defences [6], [64], perception and black-box GPS attacks on UAVs [11], [65], optimal attacks on nonlinear estimators [66], sensor reconfiguration attacks [45] and learned spoofers that exploit wind compensation [12] all establish that stealth is possible. Precise redirection with real-time tracking was demonstrated recently [67]. Acoustic injection reaches inertial sensors directly [24], [25], and the hardness of physical sensor attacks has been studied systematically [68]. None of these works treats the disguise as the attacker's optimisation variable or links undetected deviation to a mission-level certificate.

c) Detection and recovery.: Control invariants [4], SAVIOR's physical invariants with CUSUM [5], software sensors [69], PID-Piper [70], diagnosis-guided recovery [71], safety-critical variable monitoring [72], PWM-fused IDSs [73] and model-based recovery for inertial attacks [74] are the defences an attacker in our model must evade. GNSS-specific monitors include power and distortion monitoring [75] and INS-aided spoofing detection with integrity-risk bounds [76]. The latter is the closest GNSS analogue of our budget, but it has no fault or environment term and no unknown dependence. Active exposure [77] and fuzzing of inputs and parameters [78], [79] are complementary.

d) Certificates and reachability.: Randomized smoothing [7], [8], [80] and its reinforcement-learning extensions [81]–[83] certify models or rewards, not missions. Set-based reachability [84]–[86], tube model predictive control [87], funnels [88], [89], control barrier functions [90], Hamilton–Jacobi reachability [91] and the Simplex architecture [92] supply deterministic or probabilistic envelopes. Probabilistic zonotopes assume independence between their components [85]; recent work separates realised from aleatory disturbances [93]. Stochastic games give max–min reach-avoid probabilities against an adversary plus nature [94], with a fully known stochastic kernel; and a Wasserstein distributionally robust reachability metric combines stealthy false-data injection with ambiguous noise [95], without dependence between components or a mission-level floor. These two are the nearest prior work to our budget. Conformal methods [96], [97] assume exchangeability, which an adaptive attacker breaks.

e) Dependence uncertainty.: Sharp bounds on the distribution of a sum with fixed marginals [15], [16], [27], the Rearrangement Algorithm [28]–[30], [98], [99], dependency-bounds convolution [100] and probability-bounds analysis [19], [101], [102] are mature in finance and reliability engineering [103], and copulas and p-boxes appear in power-system uncertainty analysis [104]–[106]. Closest in verification are Dempster–Shafer and imprecise-copula analyses of neural networks [107]–[109] and Boole–Fréchet bounds for temporal-logic motion planning [110]. To our knowledge, no prior work applies dependence-sharp aggregation across an adversarial, a stochastic and an environmental perturbation to certify a cyber-physical mission outcome.

X. Conclusion

A UAV's navigation stack cannot tell an attacker from the weather, and an attacker who knows this will dress accordingly. We turned that observation into a threat model, a requirement and a measurement. The threat model is ε-observational equivalence, which bounds every detector's advantage by a total-variation distance. The requirement is that a mission-level guarantee must bound undetected failure without relying on the detector to notice the disguise. The measurement is a cross-layer benchmark with enforced provenance and a cross-evaluated evasion study. In that study every disguise evaded every detector, and the deviation a detector admitted depended on whether it modelled the disguise as benign, unless a trusted re-anchoring alarm capped it for all of them.

The unified budget meets the requirement. Its stochastic terms are aggregated over every coupling, including one an attacker imposes by controlling when jamming happens; its deterministic terms add no dependence gap; and its single empirical adversarial input is reported together with the case in which an independent attacker exceeded it. The analysis also corrects a hypothesis from our proposal: the union bound is not where the looseness lives. The slack lies in the dependence, envelope and temporal gaps. In our pilot the temporal gap was the largest at moderate budgets, and it is the most tractable. For the defender, re-anchoring and closing the terminal interval mattered more than any threshold.

The scaled test is the paper's other lesson. The pilot's clean result did not survive a larger test, and the violations it found came from benign missions, not attacks: from an assumption nobody monitored and from a sample asked to certify more than it could. Secured, the bound held again on fresh data, but its adversarial term is still measured, not proved. The provable alternative is $5.8\times$ looser, because the detectors that actually constrain the attacker alarm on patterns that no convex bound captures. Closing that gap, re-establishing the results in PX4 software-in-the-loop simulation, and calibrating every marginal on measured flights are the next steps, as the released runbook specifies.

Ethics Considerations

This work studies how an attacker can disguise interference with a UAV's navigation as a benign disturbance. The method is dual-use, so we weighed its benefits and harms before running any experiment, following the venue's ethics guidelines [111].

a) Stakeholders and harms.: The stakeholders are UAV operators, people and property under flight paths, autopilot maintainers and regulators. The potential harm is that an adversary uses the disguise taxonomy, or the optimiser, to make a real spoofing attack harder to detect.

b) Why the benefit outweighs the risk.:

- *The attack capability is not new.* Detector-agnostic stealthy GPS attacks against UAVs are already published [11], as are learned spoofers that exploit wind compensation [12]. Our contribution is the measurement of which disguise defeats which detector, together with a defence that does not depend on detection.

- *Every attack in this paper is paired with its mitigation in the same section:* re-anchoring-interval sizing, detector diversity across disguise classes, and the source-agnostic floor.
- *No experiment emitted radio-frequency energy.* All attacks act through simulator interfaces, and the released harness contains no transmit-capable code path.
- *No human subjects or personal data are involved.* The third-party corpora are public, are used under their licences, and are not redistributed.

c) Disclosure.: The pilot uses a generic Kalman filter rather than a specific autopilot, so it discloses no vendor vulnerability. If the software-in-the-loop transfer study in our runbook shows that a disguise family evades PX4 EKF2's innovation gating, we will notify the PX4 and ArduPilot security contacts at least 90 days before publication and include the mitigation.

Open Science

We release, through an anonymised repository for review and a permanent DOI at camera-ready [112]:

- the schema (v0.2) and its lossless adapters;
- the pairing pipeline, with mandatory provenance flags;
- the detector-operating-curve harness and the dependence-bound library (union bound, Rearrangement Algorithm, independence, comonotone);
- the simulator, the four detectors and the evasion harness, restricted to simulator interfaces;
- every result file behind every number in this paper;
- the scripts that regenerate all macros, figures and tables from those files.

Seeds are partitioned by role (calibration, optimisation, held-out attacker, held-out evaluation, negative controls). A CPU-only path reproduces every figure in about 29 minutes on two cores. Third-party corpora are fetched from their origin under their own licences. We target the Available, Functional and Reproduced badges.

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APPENDIX

Symbol	Meaning
$x, \hat{x}$	true and estimated navigation state
$e(t)$	cross-track deviation from the nominal path (m)
m	corridor margin (m); completion iff $\sup_t \lvert e\rvert \le m$
$L,\ T_c$	closed-loop Lipschitz constant; certification horizon (1 s)
$\theta,\ \theta_D$	detector operating point; threshold of detector D
$\Delta(\theta),\ s,\ H$	hidden staleness; contact-window slack; malicious hops
γ	delay-to-position rate (m/s), campaign supremum
$\delta_{\text{net}}, \delta_{\text{adv}}$	deterministic network and adversarial-offset budgets (m)
$\delta_{\text{gnss}}, \delta_{\text{fault}}, \delta_{\text{env}}$	stochastic and bounded-unknown budget terms
X_i	sup-deviation of the closed loop under source i alone
S_i	survival function of X_i, $S_i(t) = \Pr[X_i > t]$
$\underline{F}, \overline{F}$	lower and upper CDFs of a p-box ($\underline{F}$ is stochastically largest)
$\overline{p},\ p_U$	worst-case (all couplings) and union (standard) tail bounds
$f(m)$	certified floor, $1 - \overline{p}(s_m)$
U	event that the detector stack raises no alarm
T_a	re-anchoring interval (C2′)
$O_{0:T}$	defender's observations over the mission
$P_a,\ P_d$	laws of $O_{0:T}$ under attack a / disguise d
$\mathscr{D}_k$	disguise family k

Acronym	Expansion
CDF	cumulative distribution function
CEM	cross-entropy method
CMA-ES	covariance matrix adaptation evolution strategy
CUSUM	cumulative sum (sequential change detection)
EKF / KF	extended Kalman filter / Kalman filter
FPR	false-positive rate
GNSS	global navigation satellite system
IDS	intrusion detection system
IMU	inertial measurement unit
KL / KS	Kullback–Leibler divergence / Kolmogorov–Smirnov distance
LoE	loss of effectiveness (actuator)
MCR	mission-completion rate
NIS	normalised innovation squared
PD	proportional–derivative (controller)
p-box	probability box
RA	Rearrangement Algorithm
SITL	software in the loop
TV	total-variation distance
UAV	unmanned aerial vehicle

a) Proposition 2(b): sharpness for two sources.: For $d = 2$, Makarov [15] showed that $\sup_C \Pr_C[X_1 + X_2 > s] = \min\{1, \inf_t (S_1(t^-) + S_2((s-t)^-))\}$, which equals $p_U(s)$ at every continuity point of the empirical survival functions. The extremal coupling makes the events $\{X_1 > t^\star\}$ and $\{X_2 > s - t^\star\}$ disjoint, by pairing the upper tail of one source with the body of the other [27]. An attacker who controls when a stochastic source fires (for example, the timing of jamming-induced outages relative to gusts) can move the coupling towards this extremal one.

b) Proposition 2(d): comonotone and independent gaps.: Let $X_1 = \cdots = X_d = X$ almost surely, with survival function S. Then $\Pr[\sum X_i > s] = S(s/d)$, while the equal split gives $d\,S(s/d)$; whenever the equal split is optimal, the ratio is d. For two independent $\mathrm{Exp}(1)$ variables, $\Pr[X_1 + X_2 > s] = (1+s)e^{-s}$, while $\inf_t (e^{-t} + e^{-(s-t)}) = 2e^{-s/2}$. The ratio $2e^{s/2}/(1+s)$ is unbounded.

c) Monotonicity in the p-box (proof of Theorem 1).: If $X'_e \succeq_{\mathrm{st}} X_e$, there is a coupling with $X'_e \geq X_e$ almost surely. For any copula of (X_g, X_f, X_e), compose it with this monotone coupling to obtain a copula of (X_g, X_f, X'_e) under which the sum is pointwise no smaller. Hence $\overline{p}$ evaluated at the stochastically largest member of the box (CDF $\underline{F}_{\mathrm{env}}$; in the pilot, the 30 kt marginal) dominates $\overline{p}$ at every F_{env} in the box.

For a confidence level α, build an $N \times d$ matrix whose column j holds the quantiles of X_j at the levels $\alpha + (1-\alpha)\,i/N$, for $i = 1, \ldots, N$ in the upper discretisation and $i = 0, \ldots, N-1$ in the lower one. Iterate until the minimum row sum stabilises. In each iteration, rearrange every column so that it is oppositely ordered to the sum of the other columns. The resulting minimum row sum approximates the worst-case value-at-risk $\overline{\mathrm{VaR}}_\alpha(\sum X_j)$ [29], and the conservative of the two discretisations is kept. Then $\overline{p}(s) \leq 1 - \sup\{\alpha : \overline{\mathrm{VaR}}_\alpha \leq s\}$, found by bisection. The RA's convergence to the sharp bound is not proven in general. For $d = 2$, our implementation (`depbounds.py`) reproduces $\overline{p} = p_U$ to within 0.001 wherever $p_U > 0.002$; below that level, the RA's $1/N$ resolution floor dominates, which is why we report the minimum of the two bounds.

Constant	Value	Provenance
γ	1.625 m/s	parent, measured (674 samples)
s	4.84 s	parent, measured (15,392 hops)
L	1.181	parent, local estimate
step / GNSS rate	20 / 10 Hz	model
PD gains (k_p, k_d)	(1.2, 1.8)	model
drag c_d	$0.35\,\mathrm{s}^{-1}$	model
Dryden h, V	100 ft, 8 m/s	MIL-F-8785C
W_{20} envelope	15–30 kt	declared (C4)
GNSS white / GM	0.5 m / 0.8 m, 60 s	model
accel. noise / bias RW	$0.05\,\mathrm{m/s^2}$ / $0.002\,\mathrm{m/s^2}/\sqrt{\mathrm{s}}$	model (Allan class [113])
anchor σ, T_a	0.3 m, 5 s	model (C2′)
fault probabilities	0.3 / 0.2 / 0.2	model (RflyMAD categories)
outage / jam	0.3 (mean 5 s) / 0.2	model